

Crustaceans

PHYLUM	Arthropoda
SUBPHYLUM	Mandibulata
SUPERCLASS	Crustacea
CLASSES	7
ORDERS	56
FAMILIES	About 1,000
SPECIES	About 70,000

Crustaceans are a very diverse group, ranging from water fleas and copepods—only just visible to the naked eye—to heavy-bodied crabs and lobsters, which include the largest arthropods alive today. Although some crustaceans, such as woodlice, have successfully adapted to life on land, the vast majority of species live in freshwater or in the sea. Aquatic crustaceans include some of the most abundant animals on earth, and they play a key part in many food chains.

Anatomy

Crustaceans differ from insects, centipedes, and millipedes in a number of ways. They have 2 pairs of antennae, compound eyes on stalks, and a cuticle that is often strengthened with calcium carbonate, especially in larger species. The head and thorax are often covered by a shield, or carapace, and the front of this usually extends to form a projection called the rostrum. The thorax has a variable number of segments. The abdomen ends in a tail-like telson. In crabs, the abdomen is short, flat, and tightly curled around to fit under the broad carapace.

Crustacean appendages—what we commonly understand as “legs”—have 2 branches and are specialized for a number of functions such as movement, sensing the environment, respiration, and egg brooding. The first pair of thoracic legs may be enlarged to form chelipeds, with strong claws that are used for defense, handling food, and even sexual signaling. The thoracic appendages, called pereopods, typically have gills. The basal part of some appendages helps in walking, while the abdominal segments usually carry paired swimming appendages called pleopods, or swimmerets.

Feeding

Crustaceans show a wide range of feeding strategies. Most large species capture their prey and kill it by stunning, crushing, or tearing it apart. Many species are filter feeders: they use their thoracic appendages to set up currents in the water and filter small particles of food, which are amassed and pushed into the mouth. Krill feed while swimming and trap particles in hairlike structures called setae on their thoracic legs.

Small crustaceans may simply graze on particles of sediment, taking microorganisms from the surface.

Life cycle

Most species of crustaceans have separate sexes. They copulate, lay eggs, and brood the eggs, either within a brood chamber or attached to their legs. Typically, eggs hatch into tiny larvae that float away in the water, feed, and grow, developing trunk segments and legs.



RED CRAB MIGRATIONS

Christmas Island, in the Indian Ocean, has a population of about 100 million red crabs. Each year, triggered by the first rains of the wet season, they migrate from their inland habitats to mate and spawn by the coast. The males, which arrive first, dip themselves in sea water to replenish lost salt and moisture. After mating, females remain by the sea for about 2 weeks, waiting for the eggs to mature before releasing them along the coast.

INVERTEBRATES

Class Branchiopoda

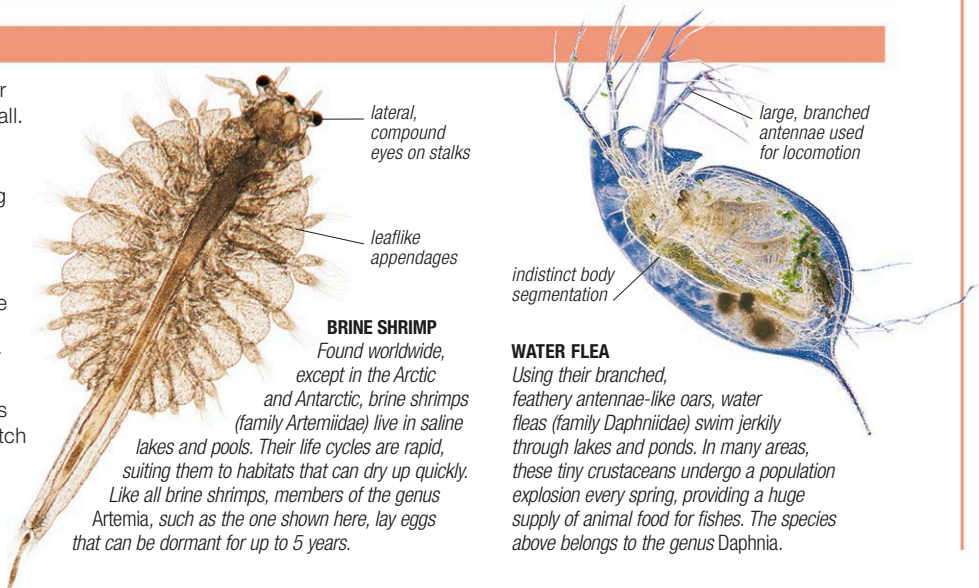
Branchiopods

Occurrence c.1,000 spp. worldwide; mainly in freshwater bodies, some in saline or marine conditions



Branchiopods comprise 4 orders: tadpole shrimps (Notostraca), water fleas (Cladocera), clam shrimps (Conchostraca), and brine shrimps (Anostraca). There are 25 families. Although some branchiopods reach a length of 4 in (10 cm), most species are much smaller. The body may be covered by a shieldlike carapace, which, in some species, takes the form of a 2-halved (bivalvate) shell. The first pair

of antennae and the second pair of mouthparts (maxillae) are small. The appendages have a flat, leaflike appearance, and carry fringes of fine bristles for filtering particles of food from the water. Branchiopods swim either rhythmically, using their appendages, or jerkily, using the second pair of antennae. Many species possess the respiratory pigment hemoglobin and may appear pink as a result. Females may produce more than one batch of eggs, each consisting of up to several hundred eggs. In some species, these are very drought resistant.



BRINE SHRIMP

Found worldwide, except in the Arctic and Antarctic, brine shrimps (family Artemiidae) live in saline lakes and pools. Their life cycles are rapid, suiting them to habitats that can dry up quickly. Like all brine shrimps, members of the genus *Artemia*, such as the one shown here, lay eggs that can be dormant for up to 5 years.

WATER FLEA

Using their branched, feathery antennae-like oars, water fleas (family Daphniidae) swim jerkily through lakes and ponds. In many areas, these tiny crustaceans undergo a population explosion every spring, providing a huge supply of animal food for fishes. The species above belongs to the genus *Daphnia*.